Deadlock

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Learning Objectives:

- Be able to explain what a deadlock is.
- Be able to read a resource-allocation graph to identify processes and resources that y could preempt to resolve deadlock.
- Be able to create a global locking order to eliminate deadlock issues.

Motivation–Dining Philosophers

- A group of N philosophers sit around a circular table.
- The philosophers want to go back and forth between:
  - Eating for some time
  - Thinking for some time
- The philosophers are eating a special dish that requires *two forks* to consume, but, the are only N forks!
- Possible solution:
  - Associate a lock with each fork.
  - Each philosopher:
    - thinks
    - acquires right fork
    - acquires left fork
    - Eats
- See `lectures/synchronization/dining.c` example. What happens when we r it?

Deadlock

- A deadlock occurs when a set of threads each waits for an event that only another thread in the set can cause.
  - It causes *starvation*, the program appears to stop executing